Evidence Debt: Measuring the Accumulated Cost of Missing Operational Evidence in Cloud-Native Systems

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***Abstract*—Technical debt names future engineering cost created by expedient technical decisions. We investigate an analogous but distinct phenomenon in cloud-native operations: *evidence debt*, the expected excess reconstruction cost created when infrastructure changes lack durable evidence of who acted, what changed, why, under which authorization, with what outcome, and how the system can be reconstructed. Missing evidence rarely causes immediate failure; cost is deferred to incidents, audits, investigations, turnover, and recovery. We contribute non-circular definitions; a reconstruction-cost model with per-source decay; analytic results showing stochastic, step-shaped "interest" and mandatory write-offs; a metric suite that rejects a scalar interest rate; an executed simulation; a bounded repository observation; and a field-study protocol. At 40% degradation, key stripping costs 0.9–16 coverage points by evidence class, while whole-record deletion collapses coverage to 32%. Joint identifier–timestamp degradation loses 35 points against a 16-point sum of single degradations; false joins reach 48% under dense activity. Reconstruction effort rises before coverage visibly fails. An age-stratified sample of 200 merged Argo CD pull requests found that 52.5% lacked an external issue reference under the declared repository schema; review-record and release-lineage presence controls were 98.5% and 100%, respectively. A seeded 50-PR manual audit matched the collector labels. This observation establishes missing external issue linkage, not inadequate intent documentation, evidence-debt magnitude, temporal decay, or reconstruction cost.**

***Index Terms*—Evidence debt, technical debt, auditability, operational evidence, cloud-native systems, DevOps, record linkage, digital forensics, site reliability engineering.**

I. Introduction

When Ward Cunningham likened expedient code to borrowed money, he gave software engineering a durable insight: some costs are not avoided but *deferred*, and deferral has a price [1]. Three decades of research have developed that insight into a family of debt constructs—code, design, architectural, documentation, and security debt among them [2, 3, 4, 5, 6]—each naming a way in which present convenience becomes future engineering cost.

This paper investigates a deferred cost that the debt family does not yet name, and that cloud-native operations manufacture at scale. When an infrastructure change is made—a deployment, a configuration edit, a permission grant, an emergency fix—an organization either captures durable evidence of the change's essential facts (who acted; what changed; why; under which authorization; with what outcome; how the resulting state can be reconstructed) or it does not. Not capturing them is almost always the locally rational choice: capture costs effort now, while the missing record costs nothing today. The cost arrives later, and in a different budget: during the incident whose responders must establish what changed last Tuesday; during the audit that must connect running workloads to authorized changes; during the security investigation that must distinguish an operator's forgotten hotfix from an intrusion; after the departure of the engineer who was the only index into a year of undocumented decisions; during disaster recovery, when "how the system can be rebuilt" turns out to live nowhere but in the system being rebuilt; and in litigation or regulatory inquiry, where the absence of records is itself a finding [7, 8].

We name the accumulated form of this deferred cost *evidence debt*, and we spend the paper making the name earn its keep. The financial metaphor is seductive and mostly wrong in its details, and one of our contributions is to say precisely where (Sections IV-E and X-A). What survives scrutiny is a phenomenon with its own structure: evidence gaps are created by identifiable practices at change time (Section II); they are syntactically observable as *deficits* against a declared evidence schema (Section III); their economic meaning is the *excess reconstruction cost* they impose on future queries against operational history (Section IV); that cost grows over time through identifiable decay processes—log expiration, ticket deletion, personnel turnover, resource ephemerality, identifier loss—whose superposition produces step-shaped, not compound, growth (Section VII); and past a knowable boundary the history becomes not expensive but *irrecoverable*, a mandatory write-off with no financial analogue (Section IV-E).

The central claim of the paper is stated by the commissioning question and treated as a hypothesis, not an assumption: *organizations can defer evidence capture, but they cannot eliminate the eventual cost of reconstructing missing operational history*. As stated, the claim is not even obviously true—redundancy across operational data sources might, in principle, make most deferred capture free, because the history could be rebuilt from what other systems recorded anyway. Our simulation study (Section VI) shows that this objection has real force at low degradation rates (redundancy absorbs single-class link loss almost completely) and fails structurally at higher rates and under record deletion, where reconstruction collapses, and—more insidiously—that heuristic reconstruction substitutes *plausible but false* history for missing history at rates that reach 48% of accepted links in our dense-activity condition. Deferred capture is thus repaid in one of three currencies: effort, error, or loss.

A. Research Questions

RQ: Can accumulated deficiencies in operational evidence be formally modeled and measured as evidence debt?

RQ1: What conditions create evidence debt?

RQ2: How can evidence debt be quantified?

RQ3: How does evidence debt influence later reconstruction effort?

RQ4: Under which conditions does evidence debt become irrecoverable rather than merely expensive?

RQ1 is answered analytically by the debt-creating-conditions taxonomy of Section II; RQ2 by the definitions, model, and metric suite of Sections III to V; RQ3 and RQ4 jointly by the model's predictions (Sections IV and VII) and their test in the executed simulation study (Section VI), with external validity addressed by the bounded repository observation and full field protocol of Section VIII.

B. Contributions and Epistemic Status

C1. Concept and definitions (conceptual): non-circular formal definitions of evidence deficit, evidence debt, principal, interest, liability, reconstruction cost, and irrecoverability, each anchored to an observable or measurable property and none defined in terms of the financial metaphor (Section III).

C2. Formal model (conceptual/analytic): evidence debt as expected excess reconstruction cost over a query workload; reconstruction as record-linkage inference over a fragmented evidence corpus [9, 10]; decay as per-source survival processes (Section IV).

C3. Anti-metaphor results (analytic): propositions showing that evidence-debt "interest" is stochastic and step-shaped rather than compound, that the debt is bounded by mandatory write-offs, and that the constructs remain well-defined with the metaphor deleted (Sections IV-E and X-A).

C4. Metric suite with rejection (methodological): six candidate metrics critically evaluated; one (a scalar evidence-debt interest rate) is rejected as ill-defined under the model and replaced by hazard-based alternatives (Section V).

C5. Executed simulation study (empirical, synthetic): a reproducible evidence-degradation experiment—complete synthetic event chains; five evidence classes degraded at 1–40%; a pairwise arm that gives the model's central interaction prediction a genuine chance to fail; removal and sensitivity arms (redundancy off, heuristic-window stress, actor-assumption breakage, single-knob density sweep) that separate design-entailed demonstrations from falsifiable tests; twenty seeds; all code and data released, with results tables generated programmatically from run outputs—including an end-to-end computation of $\mathsf{ED}(t)$ itself under a declared workload (Section VI). The study validates that the constructs are computable and discriminating; it estimates no real-world magnitudes.

C6. Public-repository observation (empirical, bounded): a reproducible age-stratified sample of 200 merged Argo CD pull requests and 30 releases under a narrow, machine-testable repository schema, with authorship/change-type strata and a seeded 50-PR manual audit. It establishes that missing external issue linkage occurs outside the simulation, while review-record and release-lineage presence controls remain discriminating; it does not measure intent adequacy, decay, debt, or cost (Section VIII).

C7. Field protocol (methodological, not executed): instruments for a deficit census, reconstruction drills, and longitudinal decay observation in real organizations (Section VIII).

Throughout, we separate established results (cited), analytic claims (proved within the model), empirical claims (measured and scoped to their source), hypotheses (labeled), and proposals (labeled). The only empirical statement about a real project is the bounded public-repository observation of Section VIII; it is explicitly separated from evidence-debt and cost measurement.

C. Scope and Non-Goals

We study operational evidence for infrastructure and configuration change in cloud-native systems. We do not propose a logging product, do not claim that more retention is always better (retention has cost and legal risk [11, 12]), and do not address the orthogonal question of whether captured evidence is trustworthy against tampering—the secure-logging literature covers integrity [13, 14, 15]; our subject is *existence, connectivity, and survivability* of evidence, which integrity mechanisms presuppose.

II. How Evidence Debt Is Created (RQ1)

Evidence debt is not created by exotic failures. It is created by ordinary, often locally rational practices at change time, and by environmental properties of cloud-native platforms that make evidence perishable by default. We analyze both, because the distinction matters for remediation: practices can be changed; platform properties must be compensated for.

A. The Evidence a Change Should Leave Behind

Fix, informally for now (formally in Definition 1), the six interrogatives a complete operational record answers for a change: *who* acted (a human or machine principal, resolvable to an accountable identity); *what* changed (the content delta, identified precisely enough to reproduce it); *why* (the intent: the problem, requirement, or decision the change served); *under which* *authorization* (the approval or policy basis that legitimized it); *with which outcome* (whether it succeeded, what it affected, what followed); and *how the resulting state can be reconstructed* (the recovery-relevant linkage from running state back to buildable sources). Contemporary toolchains capture fragments of these in different systems—tickets, commit messages, review records, CI logs, registries, deployment tools, cloud audit trails [16, 17]—keyed by correlation identifiers whose continuity is nobody's explicit responsibility [18, 19].

B. Debt-Creating Practices

The following practices, enumerated from the structure of the six interrogatives (each practice deletes or degrades at least one), create deficits at change time. We state them descriptively, not judgmentally; several are unavoidable under incident pressure, which is precisely why the debt frame—deliberate deferral with a later price—fits them.

P1: Ticketless change. Work performed without an intent record, or with intent recorded only in ephemeral channels (chat, hallway); deletes *why*.

P2: Out-of-band execution. Manual changes applied directly to runtime (console edits, `kubectl` against production, emergency patches) that bypass the pipeline where capture is instrumented; degrades *what*, *who*, and *outcome* simultaneously.

P3: Shared and impersonal identity. Changes executed under shared accounts, service credentials used interactively, or broad break-glass roles; deletes *who* even when an API trail exists [16].

P4: Approval as gesture. Authorization expressed as transient platform state (a review click, a pipeline gate) not bound durably to the executed content; deletes *under which authorization* on governance timescales.

P5: Correlation rupture. Failure to propagate identifiers across stages—no ticket reference in the commit, no commit in the build record, no digest in the deployment record; leaves all facts individually present but *unjoinable*, which Section VI shows is a distinct and subtler deficit than absence.

P6: Retention minimization by default. Accepting platform default retention (often 30–90 days for logs [12, 17, 16]) for records whose governance-relevant lifetime is years; converts captured evidence into future deficits on a schedule.

P7: Migration without evidence portability. Tool and vendor changes (forge, ITSM, CI) that migrate active data but orphan or discard historical records and their identifiers; a bulk deficit-creation event.

P8: Knowledge kept in people. Reliance on the memory of specific engineers as the de facto index into operational history; converts turnover [20, 21] into evidence decay.

C. Environmental Amplifiers

Cloud-native platforms amplify the consequences of P1–P8 in ways traditional infrastructure did not. *Ephemerality*: containers, nodes, and serverless instances are routinely destroyed and replaced [22]; state that would once have persisted incidentally on long-lived servers (shell histories, local logs, file timestamps) now vanishes by design, removing the accidental evidence that forensic practice historically relied on [23]. *Volume and automation*: reconciliation loops and autoscalers generate changes at rates no manual record-keeping can track, so anything not captured automatically is not captured [24, 25]. *Fragmentation*: the record of one logical change is scattered across six or more stores with heterogeneous schemas, identifiers, clocks, and retention policies—the precondition for the correlation failures that Section VI quantifies. *Schema and identity drift*: service renames, account migrations, and observability re-instrumentation change the vocabulary in which old evidence is expressed, so even surviving records become progressively harder to join (a decay channel distinct from deletion, Section VII).

D. Why the Cost Is Deferred, Not Avoided

Each practice above trades a certain, small, present cost (filing the ticket, propagating the identifier, exporting before migration) against an uncertain, larger, future cost that is realized only if a *query* against operational history arrives: an incident, an audit, an investigation, a recovery, a dispute. This contingent structure is what makes the debt frame apt and what makes the phenomenon invisible to steady-state operational metrics: a system can run flawlessly for years while its reconstructable history rots. Whether the deferred cost is eventually paid is therefore a question about the query workload (how often history is interrogated, with what stakes) and about decay (what survives until then)—exactly the two quantities the formal model of Section IV is built around, and the reason the central claim of Section I is a testable hypothesis rather than a tautology: if queries are rare and redundancy is high, deferral could in principle be nearly free. Section VI tests when it is and is not.

Figure 1 summarizes the lifecycle: creation at change time, silent accumulation and decay during the latency period, and realization at query time. The remainder of the paper formalizes each stage.

III. Definitions (RQ2, Part 1)

We build the definitions in strict dependency order, so that none is circular: syntactic objects first (schema, corpus, deficit), then operational quantities (reconstruction cost, irrecoverability), then the economic aggregate (debt) and its decomposition (principal, interest, liability). Each definition ends with its *observability basis*: what one would measure to instantiate it. The financial vocabulary is used only *after* the underlying quantity is defined without it.

Definition 1 (Evidence Schema)*: An* evidence schema Σ *is a declared mapping from each class of operational change to the*

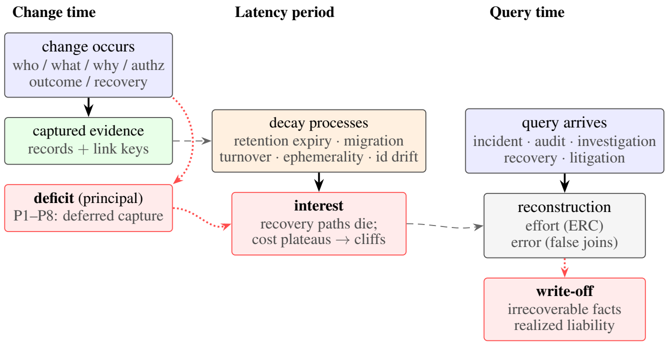

Fig. 1. The evidence-debt lifecycle. At change time, capture practices (top) either record or defer each evidence element; deferral creates a deficit (principal). During the latency period, decay processes—retention expiry, turnover, migration, identifier drift—degrade the remaining reconstruction paths (interest). The debt is realized when a query against operational history arrives (incident, audit, investigation, recovery, litigation): surviving evidence is assembled, missing links are reconstructed at effort cost, wrongly reconstructed at error cost, or found irrecoverable (write-off). Boxes name the constructs formalized in Section III.

set of evidence elements *required for that class: typed facts (actor identity, content delta, intent statement, authorization basis, outcome record, recovery linkage) together with the* link keys *that connect them across stores.* Observability: *Σ is an artifact the organization writes down; its existence is a governance choice, and all subsequent measurement is relative to it.*

The schema-relativity is deliberate and load-bearing: "missing" is undefined without a statement of what was supposed to exist. This mirrors measurement practice in data quality [26] and avoids the circularity of defining debt by whatever later turns out to be needed. The six-interrogative schema of Section II-A is our reference instantiation; organizations may declare stricter or weaker ones, and comparisons are meaningful only under a fixed Σ.

Definition 2 (Evidence Corpus)*: The* evidence corpus $\mathscr{E}(t)$ *is the set of evidence elements and link keys that durably exist and are retrievable at time t, across all stores.* Observability: *enumerable by inventory of the stores, up to discoverability limits that are themselves part of the phenomenon (Section V).*

Definition 3 (Evidence Deficit)*: For a change x of class k occurring at time t_x, the* evidence deficit $\mathsf{Def}(x,t) \subseteq \Sigma(k)$ *is the set of schema-required elements and link keys for x absent from $\mathscr{E}(t)$. The* deficit at creation *is* $\mathsf{Def}(x,t_x)$*; elements present at t_x but absent at $t > t_x$ have* decayed. Observability: *a syntactic census—for each change, check each required element's presence. No judgment about future cost is involved.*

Definition 4 (Reconstruction and Its Cost)*: A* query *q against operational history asks for a set of facts and links concerning past changes (e.g., "what changed on service s in window w, by whom, under what authorization?").* Reconstruction *is the process—human, algorithmic, or mixed—of answering q from $\mathscr{E}(t)$. Its* cost $\mathsf{C}_{\text{rec}}(q,\mathscr{E}(t))$ *is the effort expended (person-time and compute, on a declared accounting) to reach an answer of stated confidence, and its* error *is the divergence of the answer from ground truth.* Observability: *cost is measurable directly whenever reconstruction is actually performed (incident retrospectives, audit preparation, the drills of Section VIII); error is measurable wherever ground truth is independently available—natively in simulation (Section VI), partially in the field.*

Definition 5 (Evidence Irrecoverability)*: A fact or link required by query q is* irrecoverable *at t if no reconstruction procedure over $\mathscr{E}(t)$ can establish it at the required confidence —operationally, if every independent source that ever carried it has been destroyed or degraded below identifiability. A query is irrecoverable if any fact essential to it is.* Observability: *in simulation, exactly decidable (ground truth known); in the field, certified by exhaustion of an enumerated source list, which is how forensic practice certifies unavailability [27, 23].*

Definition 6 (Evidence Debt)*: Fix a schema Σ, a corpus $\mathscr{E}(t)$, and a* query workload $\mathscr{Q}$*: a set of potential future queries q with occurrence weights $w_q > 0$ (frequencies, probabilities, or scenario weights, declared explicitly). The* evidence debt *of the estate at t is the expected excess reconstruction burden induced by its deficits:*

$$\mathsf{ED}(t) \;=\; \sum_{q\in\mathscr{Q}} w_q \Big[\mathbb{E}\,\mathsf{C}_{\text{rec}}\big(q,\mathscr{E}(t)\big) \;-\; \mathbb{E}\,\mathsf{C}_{\text{rec}}\big(q,\mathscr{E}^*(t)\big)\Big] \\ \quad + \sum_{q\in\mathscr{Q}} w_q\,\pi_q\,\mathsf{Irr}_q(t),$$

where $\mathscr{E}^*(t)$ is the counterfactual ideal corpus*: every element and link key required by Σ for every change up to t, captured at change time and subject to no decay. This choice is deliberate and consequential: it attributes retention- and migration-driven loss of once-captured evidence to debt (as decay-created deficit, Definition 3), because from the query's standpoint an expired record and a never-captured record are the same absence with different histories. $\mathsf{Irr}_q(t)$ indicates that q is irrecoverable at t, and π_q is the declared penalty weight of failing to answer q at all.* Observability: *estimable, not directly observable—the estimation problem is exactly what Sections V, VI and VIII address; the definition fixes what the estimators are estimators* of.

Three features of Definition 6 do argumentative work. First, debt is *workload-relative*: an estate whose history will never be queried carries deficits but little debt; declaring $\mathscr{Q}$ forces the (often implicit) organizational judgment about which contingencies matter into the open. Second, the excess-cost form makes the baseline explicit: even a complete corpus makes queries cost something; debt is only the part deferral added. Third, irrecoverability enters as a separate, penalty-weighted term rather than as an infinite cost, keeping ED finite and decision-relevant (Proposition 4).

Definition 7 (Evidence-Debt Principal)*: The* principal *of a deficit* $\mathsf{Def}(x,t_x)$ *is its contribution to* $\mathsf{ED}(t_x)$ *at creation time—the excess expected cost were the workload queried immedi-*

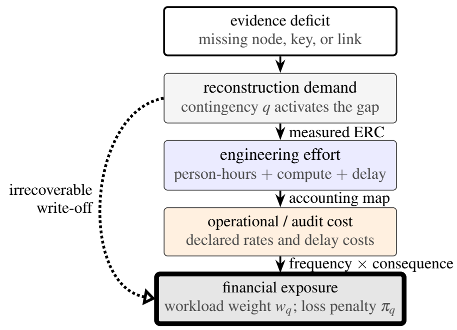

Fig. 2. The causal accounting bridge from a syntactic deficit to economic exposure. Only the first node is obtained by census. Reconstruction drills measure ERC; declared labor, compute, and delay rates map effort to cost; workload weights and consequence penalties map cost and write-offs to exposure. The figure defines an auditable chain, not a claim that this paper has estimated dollars.

ately, while memory is fresh, sources are live, and identifiers are current. Observability: *estimable at change time from the deficit census and short-horizon reconstruction calibration.*

Definition 8 (Evidence-Debt Interest)*: The* interest *accrued on change x's evidence position over* $[t_x, t]$ *is the growth of its contribution to* ED *beyond principal:* $\mathscr{I}(x,t) =$ ED-*contribution*$(x,t) -$ *principal*(x)*. It has two decay-driven components, both are interest, and they must not be conflated with new borrowing: (i)* path decay*, the loss of reconstruction routes for elements already deficient at* t_x*; and (ii)* corpus decay*, the expiry or destruction of elements captured at* t_x*, which enlarges* $\mathsf{Def}(x,t)$ *over time per Definition 3. The drivers of both are the same processes: retention expiry, record deletion, personnel departure [20], resource ephemerality [22], identity and schema drift, vendor migration, and loss of correlation identifiers.* Observability: *the decay processes are individually observable (retention policies are published [16, 17, 12]; turnover is known to HR; migrations are scheduled); Section VII shows how their superposition yields a computable cost-versus-age curve, and why it is not a compound rate.*

Definition 9 (Evidence Liability)*: The* liability *of the estate with respect to a specific contingency q at t is the conditional realized burden if q occurs at t: the actual reconstruction cost plus the consequence cost of unanswered or wrongly answered parts (regulatory exposure, prolonged outage, lost claims) [7, 8].* Observability: *realized liabilities are observable events (audit findings, incident duration attributable to archaeology);* ED *is their workload-weighted expectation, which is the standard relation between an expected cost and its realizations.*

Figure 2 makes explicit the economic bridge that the excess-cost definition otherwise compresses into one term. A deficit has no intrinsic monetary value. It becomes economically relevant only when a contingency demands reconstruction; the resulting person-time, compute, and delay can then be mapped through published accounting rates, and the workload weights and consequence penalties of Definition 6 turn those conditional costs into exposure. This separation is intentional: the paper can define and measure reconstruction burden without pretending to have measured wages, outage revenue, or regulatory loss.

The vocabulary now has the shape the brief demanded, with each term cashed out independently of the metaphor: *deficit* is syntax (what is missing), *debt* is expectation (what the missing is expected to cost), *principal* is the at-creation component, *interest* is the decay-driven growth, *liability* is the per-contingency realization, and *irrecoverability* is the boundary at which cost becomes loss. Section IV gives the machinery that connects them.

IV. A Formal Model of Evidence Debt (RQ2–RQ4)

The definitions of Section III refer to reconstruction cost and decay without structure. This section supplies the structure: operational history as a graph, evidence as a fragmented projection of it, reconstruction as linkage inference, and decay as survival processes over sources. The model is chosen so that every parameter is either directly observable or estimable by the procedures of Sections V and VIII, and so that the qualitative predictions tested in Section VI follow from it rather than from intuition.

A. History, Evidence, and Fragmentation

Ground truth. Operational history is a typed, acyclic *event graph* $\mathscr{H} = (V, L)$: nodes V are events and artifacts of a change (intent record, content revision, authorization act, build, artifact, deployment, runtime configuration, outcome), and links $L \subseteq V \times V$ are the true causal and derivational relations among them. A *chain* is the subgraph of one logical change. $\mathscr{H}$ is what actually happened; no one holds it directly.

Evidence. The corpus $\mathscr{E}(t)$ (Definition 2) is a set of *records*, each held in a store $s \in S$, each testifying to a node and carrying zero or more *link keys*: identifiers intended to name adjacent nodes (a ticket id in a commit message, a commit hash in a build record, a digest in a deployment record) [18]. The corpus thus determines an *observed graph*: nodes for surviving records, and an edge wherever a surviving key names a surviving record. Fragmentation is the model's default: records of one chain are distributed across stores with independent schemas, clocks, and retention regimes (Section II-C).

Deficit classes. Relative to schema Σ, a deficit (Definition 3) is one of: a *missing node* (the record never existed or was deleted), a *missing key* (the record exists but the link key was never propagated or was stripped), or a *degraded attribute* (e.g., a null timestamp, an unresolvable identity). The distinction is not pedantic: Section VI shows missing nodes and missing keys have different cost signatures—keys are compensable by inference at effort-and-error cost; nodes, once every copy is gone, are not compensable at all.

B. Reconstruction as Linkage Inference

A query q (Definition 4) requires recovering a subset of true links L_q. Where keys survive, recovery is lookup. Where they do not, recovery is *record linkage*: deciding, from attributes (service, actor, temporal proximity, content similarity), which surviving records refer to adjacent events—precisely the inference problem formalized by Fellegi and Sunter [9] and developed in entity resolution [10]. This identification imports the right consequences. Linkage has *effort*: candidate sets must be enumerated and compared, and effort grows with the ambient density of similar records. Linkage has *error*: with imperfect discriminators, some accepted matches are false; false links do not announce themselves, so the reconstructed history is silently wrong. And linkage has *limits*: when discriminating attributes are themselves degraded (timestamps nulled, identities recycled), candidates become indistinguishable and links unrecoverable even though both endpoint records survive.

For a single required link ℓ, let $\text{paths}(\ell,t)$ be its surviving *recovery paths* at t: the direct key, redundant keys (a digest recorded in two places), inferential linkage over attributes, and extra-corpus paths (asking the engineer who did it). Each path r has expected effort $c_r(t)$ and error rate $\varepsilon_r(t)$. A rational reconstructor uses the cheapest acceptable path, so the link's expected cost is

$$c_\ell(t) \;=\; \min\{\, c_r(t) : r \in \text{paths}(\ell,t),\ \varepsilon_r(t) \le \bar{\varepsilon}\,\},$$

with $c_\ell(t)$ undefined—the link irrecoverable (Definition 5)—when the set is empty. Query cost aggregates over required links (with shared-work economies we do not model further), and $\mathsf{ED}(t)$ follows by Definition 6.

C. Why Not $ED = f(M,F,R,U,C)$

A natural first proposal—and the one we expect readers to reach for—is a form $ED = f(M,F,R,U,C)$ over missingness, fragmentation, retention risk, uncertainty, and reconstruction cost. We considered and declined the form, keeping the factors. The reason is that the five quantities are not independent inputs to be aggregated; they occupy different roles in the mechanism. Missingness (M) parameterizes the deficit census. Fragmentation (F) and correlation uncertainty (U) shape the *cost and error of inferential paths*: fragmentation multiplies the stores to be searched, and U—driven by ambient record density and discriminator quality—sets false-match rates [9]. Retention risk (R) is a property of *decay*: it determines which paths survive to time t (Section IV-D). Reconstruction cost (C) is not an input at all; it is the *output* the others produce through the path structure. A scalar aggregation of the five would assign weights to quantities whose interaction is mediated by minimization over paths—and the minimization is what generates the phenomena that matter (plateaus, cliffs, superadditivity; Section IV-E). Any composite index built for reporting purposes (Metric M6) must therefore be understood as an estimator of Definition 6, not as its definition.

D. Decay: Survival of Recovery Paths

Each recovery path r for a link has a *survival function* $S_r(\tau)$: the probability it remains usable τ after chain creation. The empirically dominant forms are: *step survival* for policy-driven retention (a CI log is present until day T_r, then gone [12, 17, 16]); *event survival* for migrations and decommissionings (present until an organizational event at an uncertain date); *hazard survival* for human sources (available until the holder departs, with turnover hazard λ [20]); and *drift degradation* for inferential paths (discriminators weaken as identities are recycled and schemas evolve [28, 29]). Interest (Definition 8) is the consequence: as paths die, $c_\ell(t)$ jumps to the next-cheapest survivor, and when the last dies, the link crosses into irrecoverability.

E. Propositions

The following are analytic consequences of the path model; proofs are elementary and stated to make the dependency structure explicit. Their empirical counterparts are exercised in Section VI, which distinguishes design-entailed demonstrations from the falsifiable tests (notably the pairwise interaction test of Proposition 2).

Proposition 1 (Redundancy plateau)*: If a link retains at least one surviving path whose cost equals the original cheapest cost, its expected reconstruction cost is unchanged by the loss of any number of other paths. Consequently, estate-level cost as a function of a degradation rate is flat while redundancy lasts and departs from flatness only as cheapest paths begin to be destroyed.*

Proof. Immediate from $c_\ell = \min$ over surviving paths: the minimum is invariant to removal of non-minimal elements. □

Proposition 2 (Superadditive interaction of deficit classes)*: Let two degradation classes each destroy a disjoint subset of paths such that neither alone destroys any link's full path set. Then each alone adds bounded effort cost and zero irrecoverability, while their combination can produce irrecoverability—an effect strictly greater than the sum of marginal effects.*

Proof. By construction: irrecoverability requires the surviving-path set to be empty (Definition 5); if class A leaves paths $P \setminus A$ nonempty and class B leaves $P \setminus B$ nonempty, but $A \cup B \supseteq P$, the combined effect (undefined cost) exceeds any finite sum of the individual effects. □

Proposition 3 (Interest is step-shaped, not compound)*: Under the survival forms of Section IV-D, the expected cost $\mathbb{E}[c_\ell(t)]$ of a fixed deficit is a piecewise structure: constant while the cheapest surviving path persists, jumping at retention boundaries and organizational events, with smooth growth only where a hazard-governed human path is the cheapest survivor. In particular, there is no constant ρ such that cost grows as $e^{\rho\tau}$ in general: a compound "interest rate" is not a property of the process.*

Proof. Piecewise-constancy between path deaths follows from Proposition 1; jumps at step/event survival boundaries follow from minimization passing to the next survivor; the only smooth component is the expectation over hazard-distributed death times of a human path, which yields growth bounded by the next path's cost, not unbounded compounding. □

Proposition 4 (Bounded debt; mandatory write-off)*: With finite penalty weights π_q, $\mathsf{ED}(t)$ is bounded above by $\sum_q w_q(\bar{C}_q + \pi_q)$, where $\bar{C}_q$ is the cost of q's most expensive recoverable state. Once a link is irrecoverable, further decay does not increase its contribution: the debt does not grow without bound; it converts to a realized loss cap. Financial debt has no analogous mandatory write-off imposed by the world rather than chosen by a creditor.*

Proof. Directly from Definition 6: each query's contribution is at most its recoverable cost bound before irrecoverability and exactly $w_q \pi_q$ after. □

Propositions 3 and 4 are the model's verdict on the metaphor, obtained before any experiment: evidence debt has principal and carrying cost, but its "interest" is stochastic, discontinuous, and driven by third parties' retention clocks rather than by any contracted rate, and its growth terminates in write-off rather than compounding indefinitely. Section X-A completes the argument that the constructs stand without the metaphor.

Proposition 5 (Measurement reduction)*: $\mathsf{ED}(t)$ is a function of exactly four inputs: (i) the deficit census (observable, Definition 3); (ii) the path inventory with per-path cost and error calibrations; (iii) the survival schedule of paths (observable from retention policies, HR data, and migration calendars); and (iv) the declared workload (w_q, π_q). No other quantity enters Definition 6.*

Proof. By inspection of Definition 6 through Sections IV-B and IV-D: the expectation is over path survival (iii) applied to the path inventory (ii) for the links the census (i) marks deficient, aggregated under (iv). □

Two honesty notes bound what Proposition 5 claims. First, it is an input-enumeration result, not a feasibility result: input (ii)—per-path cost and error calibrations for future queries—is the empirically hard part, obtainable only by reconstruction drills (Section VIII) and bounded, not supplied, by linkage theory [9]. Second, inputs (i) and (iv) are relative to declared governance choices (Σ; w_q, π_q), so ED values are comparable across estates only under shared reference declarations (Sections V and X-B). RQ2 is therefore answered as: *quantifiable relative to declared governance inputs, with the empirical burden concentrated in cost calibration*—an affirmative with stated conditions, not an unconditional one. Section V turns the inputs into named metrics; Section VI-A.5 instantiates the entire chain, computing $\mathsf{ED}(t)$ end to end in a closed world where every input is known.

V. Metrics: Proposal and Critical Evaluation (RQ2, Part 2)

We evaluate six candidate metrics—and commit in advance to rejecting any that cannot be justified—against three criteria: *grounding* (does it estimate a quantity of Sections III and IV?), *operationalizability* (can it be computed from obtainable data?), and *incentive safety* (what does optimizing it do?). One candidate is rejected; one is accepted only with a replacement semantics; the remainder are adopted with stated scope. The foundational metrics (EDR-analogues, ERC proxies, irrecoverability, and ED itself) are computed concretely in Section VI; the replacement decay-schedule semantics is computed in Section VII; the flow metric (M4) and the composite (M6) are defined for field use and not exercised here.

Metric M1 (Evidence Deficit Ratio, EDR)*: For an estate and schema Σ: the fraction of schema-required elements and link keys absent from the corpus, reported per deficit class (missing node, missing key, degraded attribute) and per interrogative (who/what/why/authorization/outcome/recovery).* Verdict: adopted. *Grounded directly in Definition 3; computable by automated census; class-resolved reporting resists the false comfort of a single number. This is the suite's foundation metric—syntactic, cheap, and continuously monitorable.*

Metric M2 (Evidence Reconstruction Cost, ERC)*: The measured cost (person-time, compute, calendar latency) of answering a defined query from the corpus, with declared confidence, per Definition 4; collected from real reconstructions (incidents, audits) or standardized drills (Section VIII); in simulation, proxied by algorithmic effort (candidate comparisons) and wall time.* Verdict: adopted. *This is the phenomenon's direct observable; every other metric is ultimately calibrated against it.*

Metric M3 (Irrecoverability Ratio, IRR)*: The fraction of a defined query set (or of required links) irrecoverable at t per Definition 5.* Verdict: adopted. *Decidable exactly in simulation; certifiable in the field by source-exhaustion protocols [27]. Reported separately from cost metrics because it measures loss, not effort, and the two must never be averaged together (Proposition 4).*

Metric M4 (Evidence Debt Growth Rate)*: The time-derivative of debt attributable to* new *deficit creation: in practice, the deficit-census delta per unit time (new changes arriving with incomplete evidence), weighted by principal estimates.* Verdict: adopted, with renaming discipline. *This measures the inflow (are we borrowing faster?), and must be reported separately from interest accrual on the existing stock; conflating the two makes remediation invisible.*

Metric M5 (Evidence-Debt Interest Rate)*:* Proposed as*: a scalar rate ρ such that reconstruction cost grows as $e^{\rho\tau}$ or linearly at rate ρ.* Verdict: rejected. *Proposition 3 shows the growth process is piecewise, discontinuous, and schedule-driven; any fitted scalar is an artifact of the observation window and extrapolates falsely across retention cliffs.* Replacement: *the* ***decay schedule*** *of the estate—the dated sequence of upcoming path-death events (retention expirations, migrations, projected turnover quantiles) with the cost jump each implies—and, where a single summary is required, the* ***evidence half-life****: the age $\tau_{1/2}$ at which half of a reference query set ceases to be answerable at baseline cost. Both are computable from the survival inventory (Proposition 5) and are demonstrated in*

Section VII.

Metric M6 (Evidence Debt Index, EDI)*: A composite scalar summarizing estate debt for management reporting: a weighted aggregation of EDR (by interrogative), projected interest from the decay schedule, and IRR, with published weights.* Verdict: adopted with explicit estimator status and health warnings. *By Section IV-C, any such composite is an* estimator *of Definition 6 under assumptions, not a definition of it; its weights encode the workload judgment (w_q, π_q) and must be published for any comparison; and it inherits the gaming risks of all composites—optimizing EDI by weakening the schema Σ is the exact analogue of declaring bankruptcy solvent by redefining assets. EDI is defensible only alongside the disaggregated vector (EDR by class, decay schedule, IRR, measured ERC).*

Two measurement caveats apply to the whole suite. *Discoverability bias*: the census can only inspect stores it knows about; evidence in unknown stores is neither counted as present nor as deficit—field deployments should treat store discovery as part of the census. *Confidence accounting*: reconstructed answers vary in confidence; ERC must be reported at stated confidence, or costs across reconstructions are not comparable. The simulation sidesteps both (closed world, exact scoring); the field protocol confronts them directly (Section VIII).

VI. Simulation Study: Controlled Evidence Degradation (RQ3, RQ4)

We now exercise the framework in the one setting where ground truth is available by construction: a synthetic estate whose complete history is generated, then degraded under experimental control. The study is executed, not merely designed; every number below was produced by the released artifact (≈560 lines of dependency-free Python), is deterministic given the published seeds, and the results table is generated programmatically from the run outputs. Its epistemic role is stated with care, because a simulation whose apparatus is written by the authors of the model cannot "confirm" claims that the apparatus embodies. We therefore separate, explicitly, (i) *demonstrations*: behaviors entailed by design choices that mirror real toolchains, whose value is quantification and whose evidential force is tested by *removal arms* in which the choice is switched off; and (ii) *falsifiable tests*: predictions that the apparatus does not build in and that the data could have refuted. The study validates construct computability end to end—including $\mathsf{ED}(t)$ itself (Section VI-A.5)—and estimates no real-world magnitudes.

A. Apparatus

1) Ground truth

Each run generates an estate of eight services whose changes produce six-record chains across six stores—ticket (ITSM), commit (forge), approval (review), build (CI), artifact (registry), deployment (CD), plus the runtime configuration record serving as query entry point—linked by the keys real toolchains use. Six ground-truth links per chain instantiate the six interrogatives. The generator includes *redundancy* (the artifact digest recorded in both build and configuration records), mirroring practice; because this choice matters, a removal arm disables it (Section VI-C). Inter-record time gaps are drawn below the reconstructor's default heuristic window; a removal arm breaks that relationship too.

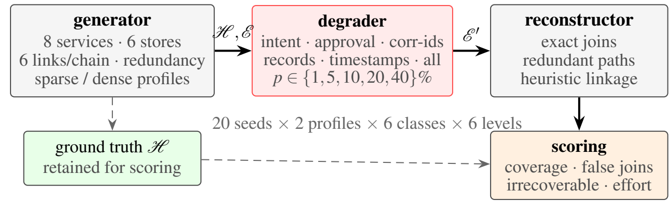

Fig. 3. The evidence-degradation experiment. A generator emits a complete synthetic estate (ground-truth chains and their records across six stores); a degrader removes a controlled fraction p of one or more evidence classes; an algorithmic reconstructor rebuilds each chain from surviving records (exact key joins, then redundant-key paths, then heuristic linkage); scoring against ground truth yields coverage, false joins, two irrecoverability constructs, invariant effort, and ED under a declared workload.

2) Factors and arms

The main factorial crosses *degradation class*—intent links, approval links, correlation identifiers, whole-record deletion, timestamps, the *pairwise* arm (correlation identifiers + timestamps, which instantiates the premise of Proposition 2: neither class alone destroys records), and an *all-classes* arm—with *level* $p \in \{0, 1, 5, 10, 20, 40\}\%$ and *activity profile* (sparse: 50 chains/service, 180 days, 12 actors; dense: 150 chains/service, 90 days, 6 actors), 20 seeds per cell. Sensitivity arms (Section VI-C) vary one design choice at a time: heuristic window below the maximum true gap; redundancy off; ticket filer $\neq$ commit author for 30% of chains; and a single-knob density sweep (chains/service $\in \{50, 100, 150\}$ with actor pool and window fixed), the latter because the sparse/dense profiles deliberately bundle three knobs and cannot isolate density.

3) Reconstructor and invariant effort accounting

For each configuration record, the reconstructor rebuilds the chain backward: surviving exact keys and redundant key paths first (implemented as index lookups and charged *zero* effort, so that effort does not depend on incidental data-structure choices), then heuristic linkage (same-service candidates within a 48-hour window, actor agreement where applicable; unique candidate accepted; multiple candidates resolved nearest-in-time—a realistic policy that can produce *false joins*). Effort is counted only at heuristic decision points, as two implementation-invariant quantities: the number of links requiring non-exact resolution (*fallback links*) and the candidate records examined there.

4) Scoring: two irrecoverability constructs

Against ground truth we score *link coverage*, *false-join rate* (incorrect/accepted), *chain coverage*, and two distinct irrecoverability constructs whose separation Section IV-A demands:

record loss (a required record exists in no store—the destroyed-source clause of Definition 5) and *identifiability-based irrecoverability* (Definition 5 in full: record destroyed, *or* no key path survives and the discriminators—timestamps, actor, service, window—fail to single out the true parent). The second is decidable here because the closed world knows the truth; it is the construct the definition means, and it behaves very differently from record loss (Finding F5). One scoring asymmetry is noted for reusers: five links are scored on the recovered parent's key, while the approval link—whose evidential record sits on the child side—is scored on the approval record's key; the artifact documents the orientation.

5) ED *under a declared workload*

To instantiate Definition 6 end to end, we declare a synthetic workload: one full-chain audit query per running configuration, unit weight; resolution cost = 1 unit per fallback link $+0.01$ per candidate examined; write-off penalty $\pi = 50$ units per identifiability-irrecoverable link. The complete corpus $\mathscr{E}^*$ resolves every link exactly, so its cost is zero and measured cost is excess cost. The declarations are arbitrary in magnitude and declared precisely because Definition 6 requires them—the point is the demonstrated pipeline from census to a computed $\mathsf{ED}(t)$, and the sensitivity of ED's composition to the declarations is itself a result (Finding F6).

B. Demonstrations, Predictions, and Results

Table I and Figure 4 summarize; released data contain all cells. Throughout, the largest per-cell coverage SD across the main factorial is 1.9 points (sparse, all-classes, p=10%).

Finding F1 (Demonstration with removal test: redundancy absorbs key loss; deletion defeats it)*: At 40% degradation of any single key class, link coverage remains at 95.3–99.1% (dense; 98.8–99.9% sparse) except correlation identifiers (84.0% dense, 96.0% sparse); whole-record deletion at the same rate collapses coverage to 32–33% with 95% of chains suffering record loss. The plateau is partly a product of the generator's (realistic) redundancy, and the removal arm shows how much: with redundancy disabled, the correlation-ids arm at p=40% (dense) falls from 84.0% to 72.6% coverage, identifiability-based irrecoverability rises from 65.3% to 84.7% of chains, and computed* ED *rises 1.6× (49.1 to 80.9 units). Redundancy is thus not an artifact to apologize for but a quantified protective factor—and its protection is bounded: no redundancy setting rescues deleted records (Proposition 1 and Definition 5).*

Finding F2 (Falsifiable test passed: the Proposition-2 pairwise arm is superadditive; the all-classes arm is deletion-dominated and subadditive)*: The pairwise arm (correlation identifiers + timestamps) instantiates Proposition 2's premise: neither class alone destroys records (record loss is 0.000 in every such cell). Timestamps alone have* zero *effect at every level in both profiles; correlation-ids alone cost 16.0 points of coverage at p=40% (dense). Their combination costs* ***35.2*** *points (dense)—more than double the sum—and* ***32.6*** *versus a sum of 4.0 points (sparse), an eight-fold superadditive excess: stripped keys force heuristic recovery exactly where nulled timestamps have destroyed the discriminators heuristics need. This is the paper's central interaction prediction, it could have failed (redundant digest paths might have absorbed the combination), and it did not. By contrast, the all-classes arm is* sub*additive in coverage relative to the sum of its five singles (e.g., 16.1 versus 16.9 points at p=5%, dense) and its losses are dominated by the deletion component (15.0 of those points): once a record is deleted, the additional loss of its keys or timestamps cannot be charged twice. Interaction structure, not headline collapse, is where the model earns or loses credit, and it behaves as Proposition 2 says: superadditivity arises between classes that disable* disjoint recovery paths*, and saturates where one class destroys the nodes outright.*

Finding F3 (Effort precedes failure; correctness outruns confidence)*: Under the invariant accounting, baseline effort is zero (every link resolves exactly). At p=40% correlation-id stripping (dense), 0.97 of 6 links per chain require heuristic resolution at 146 candidate examinations per chain, while coverage is still 84%; approval-link stripping puts 0.40 links per chain into fallback at 95.3% coverage. The estate's reconstruction economics degrade well before its dashboards would: success-rate metrics see 95%, while a growing fraction of history is being recovered by search rather than by lookup. The same cells expose a subtler and operationally important gap: identifiability-based irrecoverability at that cell is 65.3% of chains although the reconstructor's answers are 84% correct—nearest-in-time guessing is right far more often than it can* certify. *Reconstructed history under deficit is thus not merely lossy; it is unauditable even when correct, which is precisely the confidence-accounting problem Section V flags for ERC reporting.*

Finding F4 (Falsifiable test passed: density scales silent error, isolated on a single knob)*: Accepted-but-wrong links reach 14.5% (sparse) and 47.9% (dense) of accepted links in the all-classes arm at p=40%, and 61.2% (dense) under pure record deletion. Because the sparse/dense profiles bundle three parameters, the single-knob sweep varies chains-per-service alone (50/100/150; actors and window fixed) at p=*10% *all-classes: the false-join rate rises 10.1% → 16.1% → 19.5% while coverage and irrecoverability stay flat (*73.6 → 72.2%; 57.0 → 58.7%*). Ambient activity density degrades specifically the* error *channel, as linkage theory predicts [9]—and as observed in the wild in bug-link datasets, where heuristic linkage produced systematically biased corpora [30, 31]. False joins are the most dangerous repayment currency: they produce a confident, wrong account of history.*

Finding F5 (The two irrecoverability constructs diverge as the model requires)*: Record loss in deletion-bearing arms tracks the source-survival combinatorics* $1-(1-p)^6$ *within 1.2 points at every level—a consistency check that the record-loss component measures exactly what it is defined to measure (the agreement is arithmetically expected, and we claim*

TABLE I
SELECTED RESULTS, GENERATED PROGRAMMATICALLY FROM THE RUN OUTPUTS (MEAN OVER 20 SEEDS; SD IN PARENTHESES). COV = LINK COVERAGE, FJ = FALSE-JOIN RATE, IRR = IDENTIFIABILITY-BASED IRRECOVERABLE CHAINS, FB = FALLBACK LINKS PER CHAIN, ED = COMPUTED DEBT (UNITS, SECTION VI-A.5)

Class	**Prof.**	p	**Cov**	**FJ**	**Irr**	**FB**	**ED**
combined	sparse	0%	1.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.00 (0.00)	0.0 (0.0)
combined	dense	0%	1.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.00 (0.00)	0.0 (0.0)
intent links	dense	40%	0.991 (0.001)	0.009 (0.001)	0.168 (0.011)	0.40 (0.01)	9.4 (0.6)
approval links	dense	40%	0.953 (0.002)	0.047 (0.002)	0.401 (0.015)	0.40 (0.01)	21.1 (0.8)
correlation ids	dense	40%	0.840 (0.010)	0.160 (0.010)	0.653 (0.013)	0.97 (0.03)	49.1 (1.3)
timestamps	dense	40%	1.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.00 (0.00)	0.0 (0.0)
ids plus timestamps	dense	40%	0.648 (0.011)	0.180 (0.007)	0.651 (0.012)	0.86 (0.02)	47.9 (1.3)
source artifacts	sparse	40%	0.332 (0.012)	0.266 (0.020)	0.954 (0.010)	1.41 (0.04)	145.9 (4.0)
source artifacts	dense	40%	0.323 (0.009)	0.612 (0.009)	0.953 (0.008)	2.09 (0.07)	147.5 (1.8)
combined	sparse	5%	0.866 (0.015)	0.058 (0.012)	0.318 (0.025)	0.42 (0.03)	24.0 (1.9)
combined	dense	5%	0.839 (0.008)	0.145 (0.008)	0.355 (0.015)	0.44 (0.02)	27.0 (1.3)
combined	sparse	20%	0.530 (0.017)	0.137 (0.013)	0.837 (0.016)	1.25 (0.04)	97.5 (3.5)
combined	dense	20%	0.502 (0.011)	0.384 (0.010)	0.872 (0.009)	1.57 (0.05)	105.6 (2.2)
combined	sparse	40%	0.231 (0.013)	0.145 (0.019)	0.990 (0.006)	1.43 (0.04)	186.8 (3.2)
combined	dense	40%	0.234 (0.008)	0.479 (0.019)	0.992 (0.002)	1.98 (0.06)	191.2 (2.0)

nothing more from it). The construct Definition 5 actually requires is broader, and behaves differently: under pure key stripping, record loss is identically zero while identifiability-based irrecoverability reaches 65.3% (dense) and 29.7% (sparse) of chains at p=40%—history whose every record survives and whose connectivity nevertheless cannot be re-established at confidence. Timestamp nulling alone leaves both constructs at zero (keys never consult timestamps), the null result that makes the pairwise superadditivity of Finding F2 interpretable: timestamp evidence is a conditional *asset whose value is realized only when keys fail.*

Finding F6 (ED computed end to end; its composition is declaration-sensitive)*: Under the declared workload,* ED *per configuration rises from 0 at baseline to 191 units (dense, all-classes, p=40%), with the full census-to-debt pipeline of Proposition 5 exercised in code. Decomposition is instructive: at every non-trivial level the write-off term dominates the effort term by an order of magnitude or more (e.g., 51.5 of 53.5 units at p=10%, dense), i.e., under this declaration the economic mass of evidence debt sits in irrecoverable history, not in reconstruction labor. That conclusion inverts if the declared penalty π is small relative to effort costs—which is not a weakness of the construct but its content: Definition 6 makes the workload-and-penalty judgment an explicit input, and the simulation shows exactly how the judgment propagates (Section X-B).*

C. Sensitivity of the Apparatus Itself

Beyond the redundancy-removal and density-sweep arms already reported: narrowing the heuristic window to 24 hours—*below* the 36-hour maximum true inter-record gap, so that some true parents are structurally outside the search—leaves coverage nearly unchanged (83.4% versus 84.0% at the correlation-ids/dense/40% cell) while *reducing* identifiability-based irrecoverability (57.7% versus 65.3%): a narrower window excludes some true parents but disambiguates many more candidate sets. Window choice thus trades reach against confidence rather than simply scaling results. Making the ticket filer differ from the commit author in 30% of chains (breaking an actor-match assumption the generator otherwise makes perfectly reliable) lowers intent-link coverage at p=40% (dense) from 99.1% to 97.5% and raises identifiability-based irrecoverability from 16.8% to 23.4%—a measurable, modest degradation showing which headline numbers lean on that assumption. We report these arms so that readers can see the apparatus move, and the artifact accepts further policies by substitution.

D. Threats to Validity

Construct: fallback-link counts and candidate examinations are implementation-invariant but still algorithmic; human reconstruction adds search, judgment, and interruption costs—the field drills of Section VIII calibrate the mapping, and the declared unit costs of Section VI-A.5 are placeholders whose only role is demonstrating the aggregation. *Internal*: uniform-random degradation is a simplification; real decay is correlated (a migration deletes a store wholesale)—correlated-decay arms are a supported extension. The generator and reconstructor share authorship with the model; the removal and sensitivity arms are our defense, and the labeled separation of demonstrations from falsifiable tests is the honest reading of what each result can carry. *External*: the estate is synthetic; magnitudes do not transfer, and only the interaction structure—plateaus with quantified redundancy dependence, pairwise superadditivity, effort-before-failure, density-scaled error—is claimed, each tied to a model mechanism. Real-world evidence that the phenomenon and its error channel exist at scale is, however, not absent from the literature: the mining-software-repositories community measured missing bug-fix links and the bias of heuristically reconstructed histories over a decade ago [31, 30]. *Conclusion*: per-cell SDs are $\leq$1.9 coverage points; the differences supporting Findings F1, F2, F4 and F5 exceed that by an

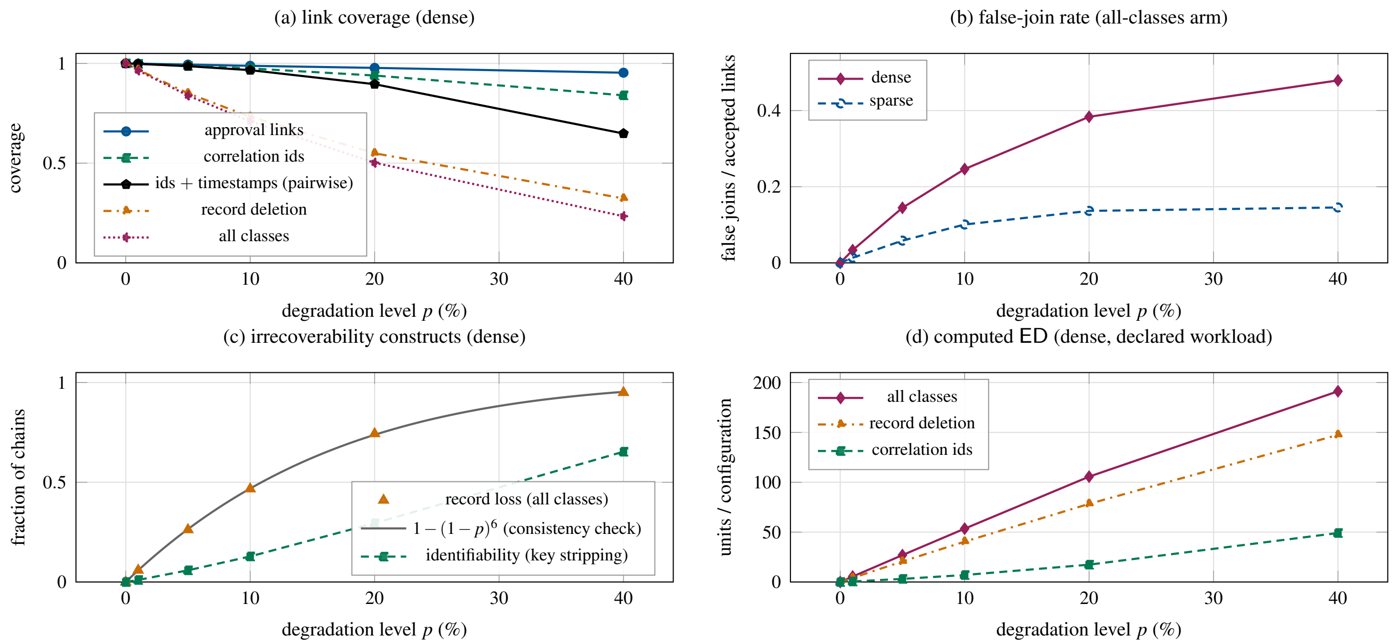

Fig. 4. Reconstruction outcomes versus evidence degradation (means over 20 seeds; SDs below marker size except where bars shown). (a) Link coverage, dense: key-stripping classes stay near ceiling; the pairwise arm (ids+timestamps) departs from the correlation-ids arm superadditively (Finding F2); record deletion and the all-classes arm collapse. (b) False-join rate in the all-classes arm: density roughly triples silent error at fixed degradation. (c) The two irrecoverability constructs, dense: record loss in the all-classes arm tracks the source-survival combinatorics $1-(1-p)^6$ (consistency check of the record-loss component, not a validation of Definition 5); identifiability-based irrecoverability under pure key stripping is large although record loss is identically zero—the constructs measure different things. (d) Computed ED under the declared workload of Section VI-A.5, dense.

order of magnitude and are stable across seeds.

VII. Interest: The Shape of Cost Growth Over Time

Proposition 3 predicts the temporal signature of evidence-debt interest: plateaus punctuated by cliffs, with smooth growth only where human memory is the cheapest surviving source. This section makes the prediction concrete by computing the expected-cost trajectory of one canonical deficit under a documented decay schedule. The computation is *model-derived and clearly labeled as such*: its inputs are realistic and cited, its output is an illustration of structure, not a field measurement.

A. A Worked Decay Schedule

Consider the paradigmatic deficit of practice P1 (Section II-B): a change deployed with no intent link—the "why" was never durably attached. Four reconstruction paths exist at creation, with survival forms from Section IV-D: the CI log, whose narrative context often reveals intent (cheap; step survival at a 90-day retention default [12, 17]); forge context—branch names, adjacent commits, review discussion (moderate cost; event survival until a forge migration, here placed at day 400, an instance of P7); the author's memory (expensive, and increasingly so with elapsed time; hazard survival with mean tenure ∼2.5 years [20]); and nothing (irrecoverable).

Figure 5 plots the resulting expected cost and irrecoverability probability against deficit age. Three features deserve narration because they are the operational content of "interest." First, *the plateau*: for 90 days the deficit is cheap to repay—this is the window in which proactive backfill (Section X-D) buys the most for the least. Second, *the cliffs*: cost multiplies at dates that are *knowable in advance*—retention expirations are published policy, migrations are scheduled projects—which is why Metric M5 replaced a fitted rate with the decay schedule itself: the schedule is actionable (extend retention, export before migration), a rate is not. Third, *the saturation*: past the last non-human path, the deficit's cost is dominated by the write-off penalty; further aging changes little because the loss has effectively already occurred (Proposition 4).

B. Evidence Half-Life

The replacement summary of Metric M5 is computed the same way: given an estate's path inventory and survival schedule, the *evidence half-life* $\tau_{1/2}$ is the age at which half of a reference query set ceases to be answerable at baseline cost (first path death), and the *irrecoverability horizon* the age distribution of last path deaths. The worked example, having a single shared schedule, makes $\tau_{1/2}$ degenerate to its one retention cliff (90 days); the summary earns its "half" only over a heterogeneous estate whose query set spans many schedules, which is its intended use. The structural point survives the degeneracy: for the paradigmatic deficit, the entire first cliff is set by a retention default that most organizations never chose deliberately [12]. That framing—reading governance-critical decay dates off ven-

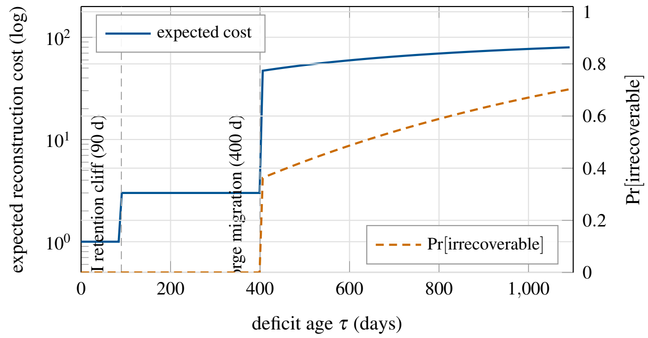

Fig. 5. Model-derived interest trajectory for a fixed intent-link deficit under the decay schedule of Section VII-A (computed from the released script; illustrative structure, not a field measurement). Expected reconstruction cost (solid, left axis, log scale) is flat while the cheapest path (CI log) survives, jumps at its 90-day retention cliff to the forge-context cost, jumps again at the day-400 migration to the interview path—whose expected cost then grows smoothly as turnover hazard accumulates and interviews lengthen—and saturates at the write-off cap as irrecoverability probability (dashed, right axis) approaches one. The shape—plateaus, cliffs, hazard-driven slope, saturation—is the general prediction of Proposition 3; none of it is compound growth.

dor defaults, rather than fitting a rate—is what makes the schedule actionable: the cheapest evidence-debt intervention is often a retention or export setting, not a new system.

C. Hypotheses for Field Confirmation

The interest analysis yields field-testable hypotheses, taken up in Section VIII: **IH1**, measured reconstruction cost for comparable queries is a step function of evidence age, with steps aligned to the estate's documented retention and migration schedule; **IH2**, reconstructions that cross a personnel-departure boundary for the relevant service cost a multiple of those that do not [20]; **IH3**, organizations' realized irrecoverability concentrates in deficits older than the last surviving automated path, with human memory as the modal final path. These are stated as hypotheses; nothing in this paper confirms them.

VIII. Repository Observation and Field-Study Protocol

The simulation validates constructs. A bounded public-repository observation can establish that named deficit classes occur outside the generator, but only reconstruction drills can establish real-world cost magnitudes and economic significance. We therefore report one executed observation, then specify the full protocol—still not executed—at implementable detail. Design follows standard guidance for case study and measurement research in software engineering [32, 33].

A. Age-Stratified Repository Observation (Executed)

We executed a read-only observation of the public `argoproj/argo-cd` repository, frozen at 2026-08-07 15:30 UTC. Selection uses `mergedAt`, not mutable `updatedAt`. Four disjoint cohorts contain 50 PRs each: the latest merges before the cutoff and the merges nearest target ages of one, three, and five years. The collector retrieves

Table II
Public Argo CD observations at the frozen cutoff

Evidence observed	Count	Share
Closing-issue link present	56/200	28.0%
Explicit non-closing issue reference only	39/200	19.5%
No external issue reference	105/200	52.5%
≥ 1 approved-review record	197/200	98.5%
Release with PR/compare reference	30/30	100.0%

Table III
Missing external issue reference by descriptive stratum

Stratum	Missing/total	Share
Recent	30/50	60.0%
≈ 1 year	28/50	56.0%
≈ 3 years	26/50	52.0%
≈ 5 years	21/50	42.0%
Human-authored	65/125	52.0%
Bot-authored	40/75	53.3%
Feature/bug	29/79	36.7%
Dependency/maintenance	48/88	54.5%
Documentation	28/33	84.8%

complete, declared date windows before sorting, so search result order does not choose the sample. We also take the 30 most recent published, non-draft releases. This is a cross-sectional age design: cohort differences may reflect changing contribution mix or project practice and cannot establish that a link disappeared over time.

Under the narrow repository schema Σ_{repo}, external issue linkage is (i) a GitHub `closingIssuesReferences` relation or (ii) a title/body token of the form `#123`, `owner/repo#123`, or a full GitHub `/issues/123` URL that resolves through the API to an Issue. Closed issues count; PR objects, unresolved/deleted targets, bare numbers, Jira keys, PR URLs, template placeholders, and references found only in comments do not. An `APPROVED` review object is only a *review-record presence control*, not proof of organizational authorization. Release-lineage presence is an explicit PR or compare reference in release notes. The executable grammar and its unit tests are released with the artifact.

The first three rows of Table II are mutually exclusive: 56 PRs retain a closing link, 39 retain only a resolved non-closing issue reference, and 105 (52.5%) retain neither under Σ_{repo}. The presence controls are high but not artificially perfect: 197/200 PRs have an approved-review record, and 30/30 releases have a change reference. Thus the collector does not simply label every public evidence channel incomplete.

Table III exposes composition rather than hiding it. Bot- and human-authored PRs have similar missing-link shares, whereas feature/bug PRs carry more linkage than maintenance or documentation PRs. The age-cohort gradient runs opposite a simple decay story: older cohorts retain more external issue linkage.

Because project norms and PR mix differ across eras, this is a descriptive warning against interpreting the observation as link decay, not evidence against the model's within-record survival mechanism.

A single-reviewer, unblinded audit of 50 PRs was sampled within cohorts using seed 20260807. Against manual title/body and resolved-target coding, the collector produced TP = 29, TN = 21, FP = 0, and FN = 0 (precision = 1.000; recall = 1.000). This validates implementation of the declared linkage grammar on the audit sample; it does not validate issue linkage as a measure of intent quality.

The observation has strict limits. A self-contained PR may document intent adequately; an issue may exist in a private or non-GitHub tracker; a dependency changelog may contribute a resolved link without documenting Argo CD intent; approval may be encoded outside GitHub; and present API state cannot prove that a link was once present and later lost. The sample is stratified and purposive, not a prevalence sample. It establishes the occurrence of missing external issue linkage under Σ_{repo}; it does not establish inadequate intent documentation, temporal decay, organizational negligence, evidence-debt magnitude, reconstruction cost, or financial exposure. Frozen data, classifications, audit coding, and generated counts are released in `empirical/`.

B. Instrument 1: Deficit Census

A static analyzer over an organization's stores (forge, ITSM, CI, registry, CD, cloud audit trail) that, for a sampled change population and a declared schema Σ, computes EDR by class and interrogative (Metric M1), the identifier continuity rate across store boundaries (P5), and the estate's decay schedule (retention settings, scheduled migrations, service-level bus factors from repository history [21]). Output: the estate's deficit and decay inventory—inputs (i) and (iii) of Proposition 5. Non-invasive; requires read access only.

C. Instrument 2: Reconstruction Drills

Timed, protocolized exercises measuring ERC directly (Metric M2): trained participants answer standardized historical queries ("for this running workload, establish what changed in window w, by whom, under which authorization, and why") using only the organization's actual stores. Recorded: person-time by subtask, sources consulted, confidence of each recovered fact, unresolved facts, and—where an independent witness exists (e.g., the original author)—error. Drills are sampled across evidence ages to test IH1–IH2 (Section VII-C) and across services with differing census scores, giving the correlation between EDR and measured ERC that a validated index (Metric M6) requires. Ethics: participants are studied with consent under institutional review; reconstruction targets are the organization's own records.

D. Instrument 3: Longitudinal Decay Observation

A yearly census re-run plus drill repetition on a fixed query panel. This observes interest directly (same query, later time, cost delta), realized irrecoverability (panel queries that become unanswerable), and the natural-experiment effects of migrations and departures as they occur. Three years covers the retention cliffs and a meaningful turnover fraction at typical rates [20].

E. Opportunistic Measurement

Real incidents and audits are unplanned drills. The protocol includes a lightweight retrospective instrument—time-to-establish-what-changed, sources used, facts never established—attachable to existing incident-review practice [34] so that realized liabilities (Definition 9) are captured at low cost. DORA-style delivery telemetry [35, 36] provides covariates (change volume, deployment frequency) for normalizing debt inflow (Metric M4).

F. What Would Refute the Framework

The framework fails empirically if: census EDR does not predict drill ERC beyond chance (the syntactic layer is disconnected from the economic one); measured cost curves are smooth in evidence age with no alignment to retention/migration schedules (against IH1 and Proposition 3); or false-fact rates in drills are negligible even under heavy deficit (against Finding F4's transfer). Each refutation would be informative, and the instruments are designed to detect them.

IX. Related Work and the Novelty Test

A. The Debt Family

Technical debt began as a metaphor for expedient code [1] and matured into a research field with theory, taxonomy, and management practice [2, 3, 4, 5, 37, 38]. The family has since acquired members—architectural, test, documentation, security, social, and hidden machine-learning debt [5, 6, 39, 40, 41]—and, recently, organizational debt [42]; practitioner literature gestures at "operational debt" [43] and, closest in spirit to our subject, "dark debt": deferred, invisible operational cost that reveals itself only in anomalies [44]. Evidence debt gives one species of dark debt a measurement theory. Evidence debt shares the family's defining structure (present expedience, deferred cost, accruing carrying cost, principal/interest decomposition) and differs in its *object*: every existing debt type is a property of the *system or organization as it is*—its code, architecture, tests, prose, security posture, structure—whereas evidence debt is a property of the *record of how it came to be*. The distinction has consequences the family's tools cannot absorb: evidence debt is invisible to any inspection of the current system state (a flawless codebase can carry catastrophic evidence debt); its carrying cost is driven by third-party retention clocks and personnel turnover rather than by ongoing development friction; and it uniquely exhibits mandatory write-offs (Proposition 4)—no amount of later diligence recovers a record that no longer exists anywhere, whereas any code can, at some price, be refactored.

Self-admitted technical debt [45] offers an instructive contrast: there, the record (a TODO comment) survives and the remediation is deferred; here, the remediation (the system works) is complete and the *record* is what was deferred.

B. Documentation Debt

The nearest family member deserves its own confrontation. Documentation debt names missing or outdated *descriptions of the system*: design docs, API references, runbooks [5]. Evidence debt names missing *records of events*: who changed what, when, why, under what authority, with what outcome. The two differ in referent (state versus history), in falsifiability (documentation can be regenerated from the system it describes; events cannot be re-observed once their traces are gone), and in decay mechanics (documentation rots by divergence from a living system [28, 29]; evidence rots by destruction of sources and loss of joinability). The software-maintenance tradition has long measured the cost of understanding systems whose rationale is lost [46, 47]—"software archaeology" is reconstruction effort by another name, applied to code rather than operations. A team with perfect current documentation and no change history has zero documentation debt and maximal evidence debt.

C. Observability and Logging

Observability practice [48, 19, 18] and log management [12] concern the capture and interrogation of operational telemetry, and are the mechanical substrate on which evidence capture rides; empirical studies show that even this substrate is captured incompletely and inconsistently in practice [49]. But observability's design center is the *present*: understanding the system's current behavior, with retention windows sized for debugging, not governance. The obvious objection—"is this not just missing logs?"—is answered by three structural gaps. First, logs are one store among six-plus; the deficits that dominated our experiment—broken intent and authorization linkage, correlation rupture—live *between* stores, and no log completeness fixes joinability (Finding F2). Second, missing logs is a state description; evidence debt is an economic quantity over a query workload with decay dynamics (Definition 6)—one can have complete logs and high debt (all short-retention) or gappy logs and low debt (rare, low-stakes queries). Third, the observability literature has no counterpart of irrecoverability, interest schedules, or principal—the constructs that make deferral decisions analyzable. Secure-logging research [13, 14, 15] guarantees integrity of what is kept and is orthogonal, as scoped in Section I.

D. Provenance, Forensics, and Records Management

Provenance systems capture derivation graphs [50, 51, 52]—when deployed, they are evidence-debt *prevention* infrastructure, and our evidence graph (Section IV-A) is deliberately PROV-compatible. What the provenance literature does not provide is the economics of its own absence: what it costs to *not* have deployed it, which is this paper's subject. Digital forensics [23, 27] supplies the reconstruction craft and the source-exhaustion standard our irrecoverability certification borrows; its setting is adversarial and case-bound, whereas evidence debt is an estate-level, mostly non-adversarial accounting. Records management and archival science [53, 54, 55] have long managed retention against future obligation; evidence debt can be read as bringing their concern into software operations with a formal cost model attached, where change velocity and fragmentation are orders of magnitude beyond the archival setting.

E. Knowledge Management and Turnover

Organizational memory research established that organizations forget through personnel flow [56, 57], and empirical software engineering has quantified turnover-induced knowledge loss [20] and truck factors [21]. In our model these appear precisely as the hazard survival of human recovery paths (Section IV-D)—the term that makes interest smooth between cliffs (Figure 5). The debt frame adds what the knowledge literature lacks for this domain: a per-change, schema-relative, countable unit of deferral (the deficit) connecting individual operational decisions to the aggregate forgetting rate.

F. Linkage, Traceability, Process Mining, and Data Quality

Our reconstruction theory imports record linkage [9, 10] (effort/error structure of inferential joins), the missing-data tradition [58, 59] (missingness as a mechanism, not just a rate—operational evidence is emphatically not missing at random: P1–P8 delete specific interrogatives under specific pressures, a bias field studies must model), and data-quality theory [26] (fitness-for-use relativity, mirrored in our schema- and workload-relativity).

Three empirical traditions have, in fact, already measured instances of the phenomenon this paper names, and confronting them sharpens both the novelty claim and its limits. The mining-software-repositories community documented *missing links* between bugs and fixing commits—our P5, correlation rupture, in the wild—and showed that heuristically reconstructed link sets are not merely incomplete but systematically *biased*, corrupting downstream conclusions [31, 30]: field evidence that the false-join channel of Finding F4 operates outside simulations. Requirements-traceability research has treated missing artifact links and their after-the-fact recovery for three decades [60, 61], with the same effort/precision economics. Process mining reconstructs operational processes from partial, noisy event logs and maintains an explicit log-quality literature [62]. What none of these supplies—and what evidence debt adds over them—is the estate-level economic aggregation: a schema-relative deficit census, decay dynamics with principal/interest structure, an irrecoverability boundary, and a workload-weighted cost model connecting today's capture decisions to tomorrow's reconstruction burden. Conversely, these literatures ground our claim that the phenomenon is real and measurable at scale; they are the natural execution venues for parts of Section VIII.

Configuration drift and IaC research [63, 24, 25] concern divergence of running state from desired state; evidence debt concerns divergence of *recorded history* from *actual history*—a system with zero configuration drift can be maximally evidence-indebted about how it got there. Insecure IaC practice itself illustrates deficit creation at the source [64]. Supply-chain

	Technical debt	Evidence debt
object	system as it is (code, design)	record of how it came to be
visible via	inspection of current system	only via queries against history
interest driver	ongoing development friction	retention clocks, turnover, migration
growth shape	roughly continuous	plateaus and cliffs (Prop. 3)
terminal state	always refactorable, at a price	mandatory write-off (Prop. 4)
repayment currency	engineering effort	effort, then error, then loss

Fig. 6. Technical debt versus evidence debt. Both share the deferral structure (left): an expedient act now, a carrying cost, a repayment event. They diverge in every load-bearing property (right): object, visibility, interest driver, growth shape, terminal state, and repayment currency. The comparison is the condensed form of Section IX-G.

integrity work [65, 66] makes particular evidence chains cryptographically strong; it hardens what is captured rather than pricing what is not. Regulatory control catalogs demand auditable change records outright [67, 8]; evidence debt supplies the quantity those obligations implicitly assume is being managed.

G. The Novelty Test, Answered Directly

Why is this not simply technical debt? Different object (record of history versus state of system), different decay drivers (third-party clocks and turnover versus development friction), different terminal behavior (mandatory write-off versus indefinite refactorability), different observability (invisible to system inspection). *Why not documentation debt?* State versus events; regenerability versus unrepeatable observation (Section IX-B). *Why not missing logs?* Logs are one store; the phenomenon is cross-store joinability, workload-relative cost, and decay dynamics; completeness of capture at one point neither implies nor is implied by low debt (Section IX-C). *Why not observability debt?* Insufficient observability is one *creation condition* among the eight of Section II-B; the debt also grows from retention, migration, turnover, and identifier rupture after perfectly observable capture. *What unique phenomenon does evidence debt capture?* The deferred, decaying, workload-contingent cost of reconstructing operational history—a quantity with its own creation conditions (RQ1), estimator theory (RQ2), effort-before-failure dynamics (RQ3, Finding F3), and irrecoverability boundary (RQ4, Findings F1 and F5)—none of which is expressible as a property of the current system state that the existing debt family describes.

Figure 6 condenses the comparison. The claim, precisely bounded: evidence debt is a member of the debt family by structure, and reducible to no existing member by object, dynamics, or management levers.

X. Discussion

A. Attacking the Metaphor, Deliberately

A concept paper owes its readers the strongest case against itself. We therefore attack our own framing and report what survives. *Where the financial analogy fails:* there is no creditor and no contract—nobody agreed to the terms, and the "lender" (future circumstance) never negotiates; the interest is not a rate but a stochastic schedule of cliffs and hazards (Proposition 3), so compounding intuitions—and any metric built on them—mislead (Metric M5 was rejected on exactly this ground); repayment is not fungible—effort, error, and loss (Findings F1, F3 and F4) are different currencies that do not net against each other; and write-offs are mandatory and world-imposed (Proposition 4), where financial write-offs are creditor choices. *Where the analogy earns its keep:* the deferral structure is real (a present choice with a later, larger, contingent cost); the principal/carrying-cost decomposition marks a real operational distinction (capture-time gaps versus decay-driven growth) with distinct remedies (capture discipline versus retention and export policy); and "debt" correctly imports the management insight that the stock must be measured and budgeted, not merely deplored.

The removal test: delete the word "debt" and every construct still stands—deficit (Definition 3) is a syntactic census; reconstruction cost (Definition 4) is a measured quantity; decay schedules and half-lives (Section VII) are computed from published retention and turnover; irrecoverability (Definition 5) is a decidable/certifiable state; and the expected-excess-cost aggregate (Definition 6) is an ordinary expected value that could as well be called *reconstruction exposure*. We embrace that reading: the *measured* quantity is reconstruction exposure, and a skeptic may use that vocabulary throughout without loss of formal content.

What, then, does the debt framing add that exposure does not? Exposure language frames the quantity as environmental—a risk one has, like weather. The debt framing makes one further, contentful commitment: it binds the aggregate to *identifiable, dated, attributable deferral decisions*—this change, shipped ticketless, under this pressure, by this team—and thereby licenses management machinery that exposure language does not naturally carry: per-decision registration at creation time (the register of Section X-D, with backfill due dates set inside the cheapest-path window); attribution of the aggregate to controllable practices P1–P8 rather than to fate; and the inflow/stock separation (Metric M4 versus interest) that tells an organization whether it is borrowing faster or merely aging. These are decisions and accountabilities, not measurements; the measurement structure is metaphor-free, and the metaphor's residual work is to keep the human origin of the exposure attached to the number. That is a naming choice we defend as useful—while conceding, per the analysis above, that every quantitative intuition imported from finance (rates, compounding, refinancing) must be checked against the model and mostly fails.

Circularity was the other standing risk, and the definitional order of Section III was built against it: deficit is defined without reference to cost (syntax against a declared schema); cost is defined without reference to deficit (measured effort of answering queries); debt relates the two through an explicit counterfactual; and the schema- and workload-relativity are declared inputs rather than smuggled conclusions. What *is* irreducibly judgmental —the choice of Σ and (w_q, π_q)—is surfaced as a governance decision, not hidden in a formula (Section IV-C).

B. Relativity of the Construct

The sharpest structural limitation is not the metaphor but the relativity: ED is defined against a declared schema Σ and a declared workload (w_q, π_q), both controlled by the party being measured. An estate can reach $\mathsf{ED} \approx 0$ by declaring a weak schema or a sparse workload, and Finding F6 showed concretely how the penalty declaration reshapes what the number says. We therefore state plainly: what is measurable *intersubjectively* is the ingredient vector—EDR against a published schema, measured ERC at stated confidence, the decay schedule, and the irrecoverability constructs—while ED itself is an accounting quantity relative to published governance inputs, comparable across estates only under shared reference declarations. To make comparison possible rather than merely honest, we propose (as future standardization, not as this paper's result) a *reference declaration*: the six-interrogative schema of Section II-A as Σ_{ref}, and a workload $\mathscr{Q}_{\text{ref}}$ parameterized by public covariates—per-service audit queries at a fixed annual rate, incident queries at the estate's measured incident rate, one recovery query per service—with penalty weights published alongside any reported figure. This is the same discipline financial accounting imposes through reporting standards: the number is meaningful because the rules for producing it are shared, not because it is observer-independent.

C. Explanatory and Predictive Value

The concept must be useful beyond naming. Its explanatory content: it unifies under one mechanism phenomena currently discussed separately—why incidents on old systems take longer to diagnose (interest), why audits of healthy systems fail (deficit despite health), why tool migrations feel like organizational amnesia (bulk path death), why the departure of one engineer disproportionately damages some services (human path as last survivor). Its predictive content is concrete and falsifiable: cost steps aligned to retention/migration schedules (IH1), departure-boundary cost multiples (IH2), memory as modal final path (IH3), and the simulation-grounded expectations that effort inflation precedes failure and that dense, fragmented estates pay more in false history than in missing history (Findings F3 and F4). Section VIII specifies the refutation conditions; a concept that survives them will have earned its place, and one that does not will have been usefully eliminated.

D. Management Implications (Proposals)

Stated as proposals, not findings. *Evidence-debt register*: like a TD register [37, 38], an explicit ledger of known deficits with principal estimates and decay deadlines—P2 break-glass changes, for instance, entered at creation with a backfill due date inside the cheapest-path window (Section VII-A). *Retention as governance, not default*: Section VII-B's observation that half-lives are set by unexamined vendor defaults makes retention review the highest-leverage, lowest-cost intervention. *Backfill sprints as refinancing*: deliberate reconstruction of high-value history while paths are cheap—though Finding F4 warns that late backfill in dense estates risks writing confident fiction into the record, so backfilled facts should carry confidence annotations. *Deferral budgets*: if evidence capture is to be skipped under pressure, the skip should be a recorded decision—the debt family's central lesson [2] transposed: unmanaged debt is not the debt you took on but the debt you never knew you took on.

E. Limitations

Beyond the simulation threats (Section VI-D): the framework prices reconstruction, not the first-order value of evidence for real-time operations (alerting, debugging), which observability already treats [48]; the workload $\mathscr{Q}$ is a declared judgment and organizations may declare it self-servingly—the discipline is publishing it; capture itself has costs (effort, storage, privacy exposure under data-protection law [11, 12]) that our model takes as given rather than optimizing, so the full decision problem—optimal capture under cost and legal constraint—remains open; and the only non-synthetic observation is a bounded, age-stratified public-forge sample that shows missing external issue linkage occurs under a declared schema but measures neither intent adequacy, decay, reconstruction, nor money (Table II). All cost, error, interaction, and debt magnitudes remain synthetic, a boundary marked at every claim.

XI. Data and Code Availability

The simulation code, canonical synthetic outputs, repository-observation collector, frozen age-stratified sample, manual-audit coding, generated tables, and reproduction instructions are archived at Zenodo as version 1.2.0, DOI 10.5281/zenodo.21843147; the version-independent concept DOI is 10.5281/zenodo.21841440. The development repository is `github.com/obedebessa/evidence-debt`; the archival source is tagged `v1.2.0`. The file `MANIFEST.sha256` records a SHA-256 digest for every released artifact, and `CITATION.cff` records the version-specific DOI. The frozen observation can be verified offline; live recollection is separated because public repository state can change.

XII. Conclusion

Cloud-native operations have made change cheap, fast, and voluminous—and have thereby made the *record* of change the system's most perishable component. This paper named the resulting deferred cost evidence debt and did the work the name requires: non-circular definitions grounded in observables; a formal model in which debt is expected excess reconstruction cost, reconstruction is linkage inference, and decay is a survival process; analytic results locating exactly where the financial metaphor breaks (stochastic step interest, mandatory write-offs) and what remains when it is removed; a metric suite honest enough to reject one of its own commissioned members; an executed, fully reproducible degradation experiment showing that redundancy absorbs small deficits, that deficit classes collapse reconstruction superadditively, that effort inflates before failure appears, and that reconstruction under heavy deficit fabricates confident false history at rates that should alarm anyone who has trusted an after-the-fact timeline; a bounded, age-stratified Argo CD observation showing that missing external issue linkage occurs in public repository evidence while review-record and release-lineage presence controls remain high; and a field protocol with stated refutation conditions. The repository observation is not a measurement of intent quality, decay, evidence debt, or reconstruction cost.

The central claim—deferral without elimination—emerges from this treatment with its proper status: not an axiom but a structured hypothesis, true in the simulation's closed world in a specific, mechanistic way (payment in effort, then error, then loss), and awaiting field magnitudes. The constructs are offered for reuse: a deficit census any organization can run, a decay schedule any organization can read off its own retention settings, and a definition of debt that makes the implicit question—*what will it cost us to not know what we did?*—explicit, estimable, and budgetable. Systems will keep forgetting; the choice organizations retain is whether the forgetting is priced.

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